Native cut-aware norm representation profile

Hypothesis: at least one bounded affine support limit reduces actual native internal coordinates by10% versus the exact limit1 reference. This is a representation-count prediction, not a speedup claim.

The same174-primary-bit actual GF(2^83),d18 norm predicate was used under every arm. The source, fixture hash, seven limits and150000 preclean-coordinate admission estimate cap were frozen before this target profile. FourGiB, CPU0 and90-second outer budget were retained. No SAT search or Gaussian initialization ran.

Independent controls passed73216 literal polynomial-norm assignments,116480 hybrid complete-primary assignment checks, and seven native model sets of28 models each. Unique functional extension gives a bijection on primary models. Constants, zero and repeated sources are retained.

Recorded arms:

limit 1: external 58176, XOR rows 44921, actual native internal 58176.

limit 2: external 40148, XOR rows 26893, actual native internal 45427.

limit 4: external 30509, XOR rows 17254, actual native internal 55395.

limit 8: external 26389, XOR rows 13134, actual native internal 74121.

limit 16: external 23754, XOR rows 10499, actual native internal 106094.

limit 32: external 22121, XOR rows 8866, actual native internal not loaded: preregistered cap.

limit dense: external 16134, XOR rows 2879, actual native internal not loaded: preregistered cap.

Worker status: passed. Profile complete: True. Preregistered10% prediction met: True.

The native count includes cut coordinates created after explicit pending-addition flush. Raw coefficient counts and preclean estimates remain in the result. A skipped dense arm is capped, not an empty root set or an UNSAT result. No original-root count, timing gain, new field-family performance coverage or novelty is claimed.

Decision and next experiment: native-hybrid-initialization-and-new-hard-search-v2. Retain every arm, use full native matrix statistics and literal witness replay before hard search, and keep original S3 ghost/exception/auxiliary coverage and the matching genuine F4/F5/F6 eightfold end-to-end goal unresolved.

Novelty review: the author CryptoMiniSat documentation and the NIST Boyar/Matthews/Peralta2008 record establish earlier XOR/circuit-minimization work. PRIMARY_SOURCES.md limits this review to opened documentation and abstract; no new priority is claimed.

Evidence: research/hybrid_norm_xor_20261017_v1/ and lab/ledger/attempts/native-cut-aware-hybrid-

norm-profile-n83-20261017-v1/0001/. Canonical performance graphs remain unchanged because this is not a qualified end-to-end result. New report publication remains pending the earlier separate authorization; source code was committed to PR2049.

